

CSE-439s

Design of Compilers

Project Test Snapshots

Team #24

Names	ID
Mathew Mina Youssef	2300910
Martin Amgad Zakaria	2200751
Radwa Yasser Ahmed	2200239
Hanya Ibrahim Aref	2201045
Nouran Mohamed El-Sayed	2201382
Andrew Ehab Tharwat	2200249

Test Case 1

Input

```
read x;
if x < 10 then
  y := x + 2 * (3 - 1) / 4;
  write y
end
```

Output

```
read, READ
x, IDENTIFIER
;, SEMICOLON
if, IF
x, IDENTIFIER
<, LESSTHAN
10, NUMBER
then, THEN
y, IDENTIFIER
:=, ASSIGN
x, IDENTIFIER
+, PLUS
2, NUMBER
*, MULT
(, OPENBRACKET
3, NUMBER
-, MINUS
1, NUMBER
), CLOSEDBRACKET
/, DIV
4, NUMBER
;, SEMICOLON
write, WRITE
y, IDENTIFIER
end, END
```

Test Case 2

Input

```
read x;
if 0 < x then
  fact := 1;
  repeat
    fact := fact * x;
    x := x - 1
  until x = 0;
  write fact
end
```

Output

```
read, READ
x, IDENTIFIER
;, SEMICOLON
if, IF
0, NUMBER
<, LESSTHAN
x, IDENTIFIER
then, THEN
fact, IDENTIFIER
:=, ASSIGN
1, NUMBER
;, SEMICOLON
repeat, REPEAT
fact, IDENTIFIER
:=, ASSIGN
fact, IDENTIFIER
*, MULT
x, IDENTIFIER
;, SEMICOLON
x, IDENTIFIER
:=, ASSIGN
x, IDENTIFIER
-, MINUS
1, NUMBER
until, UNTIL
x, IDENTIFIER
=, EQUAL
0, NUMBER
;, SEMICOLON
write, WRITE
fact, IDENTIFIER
end, END
```

Test Case 3

Input

```
{ Sample program in TINY language -
computes factorial}
read x; {input an integer }
If 0<x then { don't compute if
x <= 0 }
fact := 1;
repeat
fact := fact * x;
x := x - 1
until x = 0;
write fact { output factorial of
x }
end
```

Output

```
read, READ
x, IDENTIFIER
;, SEMICOLON
If, IF
0, NUMBER
<, LESSTHAN
x, IDENTIFIER
then, THEN
fact, IDENTIFIER
:=, ASSIGN
1, NUMBER
;, SEMICOLON
repeat, REPEAT
fact, IDENTIFIER
:=, ASSIGN
fact, IDENTIFIER
*, MULT
x, IDENTIFIER
;, SEMICOLON
x, IDENTIFIER
:=, ASSIGN
x, IDENTIFIER
-, MINUS
1, NUMBER
until, UNTIL
x, IDENTIFIER
=, EQUAL
0, NUMBER
;, SEMICOLON
write, WRITE
fact, IDENTIFIER
end, END
```